

FEmmG-FHE: Fibonacci-Golden Ratio Cryptography

An Exploration of the φ -Extension Ring for
Noise Management and Program Encoding in Lattice-Based Cryptography

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Abstract

We explore the algebraic extension ring $R[X]/(X^2 - X - 1)$ where R is a cyclotomic polynomial ring, as a primitive for lattice-based cryptography. The golden ratio $\varphi = (1 + \sqrt{5})/2$ and its conjugate $\psi = -1/\varphi$ serve as the two roots of the extension, enabling a decomposition $R[X]/(X^2 - X - 1) \cong R \times R$ via the Chinese Remainder Theorem.

We investigate three applications of this structure:

Noise Management for FHE: The φ/ψ split separates signal and noise into distinct algebraic directions. A “clean” operation using only homomorphic addition (zero ciphertext multiplication depth) attenuates the ψ -component by a factor of $|\psi|^2 \approx 0.382$ per cycle. Combined with standard CKKS bootstrapping, this enables extended computation depth. We demonstrate 6,297 sequential homomorphic multiplications across 99 bootstraps. We observe that error growth is exponential with a very small base (~ 1.00075 per multiplication), not linear as we initially hypothesized. Projecting this rate yields approximately 14,000 multiplications to 1% error—a practical improvement over standard CKKS but not “unlimited” depth.

Dual-Reality Program Encoding: The two simultaneous “realities” (φ and ψ) can encode two functionally equivalent representations of a computation. An observer seeing both outputs cannot determine which representation was the intended one (51.2% adversary success rate, 95% CI: $50\% \pm 3.1\%$). We explicitly note this is **not** indistinguishability obfuscation in the standard cryptographic sense [8, 9]; it provides plausible deniability for program representation rather than general circuit hiding.

Compact Key Encapsulation: Storing φ and ψ evaluations instead of full polynomial coefficients yields small key sizes. Our experimental variant achieves 192 bytes total. We emphasize this uses Ring-LWE with a fixed public matrix—a different and less-studied assumption than the Module-LWE used by NIST-standardized Kyber. The size comparison to Kyber-1024 is provided for scale reference, not as a claim of equivalent security.

All experiments were conducted on consumer hardware (AMD Ryzen 5 2600, 16GB RAM, WSL2 on Windows 11). Code is available at <https://github.com/primordialomegazero/femmgFHE>. We have sought to document all limitations honestly and welcome independent verification and critique.

1 Introduction

Fully Homomorphic Encryption (FHE) enables computation on encrypted data without decryption. Since Gentry’s breakthrough [1], FHE has advanced through multiple generations [3–5]. Despite these advances, noise accumulation during homomorphic operations remains the central challenge.

Motivation. We became interested in the ring $R[X]/(X^2 - X - 1)$ because its two roots have an unusual property: one is greater than 1 ($\varphi \approx 1.618$), the other is less than 1 in absolute value ($|\psi| \approx 0.618$). This suggested the possibility of asymmetric dynamics—amplification in one algebraic direction, attenuation in another.

What We Found. Our exploration yielded three observations:

1. A noise attenuation technique for FHE that uses only homomorphic addition (zero multiplication depth), enabling extended computation chains.
2. A dual-reality encoding where two equivalent program representations produce outputs that are difficult to distinguish without the secret key.
3. A key compression method that stores ring element evaluations rather than full coefficient vectors.

What We Did NOT Find. We want to be clear about the limits of this work:

- We have **not** achieved linear error growth. Our initial claim was incorrect; re-analysis shows exponential growth with base ~ 1.00075 .
- We have **not** constructed indistinguishability obfuscation in the standard cryptographic sense. Our dual-reality encoding provides plausible deniability, not general circuit hiding.
- We have **not** proven security equivalent to NIST-standardized KEMs. Our construction uses different assumptions.
- We have **not** obtained third-party verification. All results are author-reported.

Paper Organization. Section 2 establishes the mathematical foundation. Section 3 describes the FHE noise management technique. Section 4 presents the dual-reality program encoding. Section 5 details the compact KEM. Section 6 surveys related work. Section 7 discusses limitations. Section 8 concludes.

2 Mathematical Preliminaries

2.1 The Golden Ratio Extension Ring

Definition 2.1 (φ -Extension Ring). *Let $R = \mathbb{Z}_q[X]/(X^N + 1)$ be a cyclotomic polynomial ring. The φ -extension ring is:*

$$R_\varphi = R[Y]/(Y^2 - Y - 1) \tag{1}$$

where $Y^2 = Y + 1$. The two roots are $\varphi = (1 + \sqrt{5})/2 \approx 1.618$ and $\psi = -1/\varphi \approx -0.618$.

Theorem 2.2 (CRT Decomposition). *If $\varphi - \psi$ is invertible in R , then $R[Y]/(Y^2 - Y - 1) \cong R \times R$ via $a + bY \mapsto (a + b\varphi, a + b\psi)$.*

Proof. $Y^2 - Y - 1 = (Y - \varphi)(Y - \psi)$ splits into distinct linear factors. Since $\varphi - \psi = \sqrt{5}$, invertible when $q \nmid 5$, CRT applies directly. $\square$

Every element $v = a + bY \in R_\varphi$ has two “evaluations” $\text{eval}_\varphi(v) = a + b\varphi$ and $\text{eval}_\psi(v) = a + b\psi$. These are independent; knowing one does not determine the other without the secret key.

2.2 Zero-Depth Operations

The operation $\text{mul}_Y(a, b) = (b, a + b)$ corresponds to multiplication by Y . It requires only one addition and two copies—zero ciphertext multiplication (**EvalMult**) calls. Its inverse is $\text{div}_Y(a, b) = (b - a, a)$, requiring one subtraction.

Proposition 2.3 (Eigenvalues). *The matrix representation of mul_Y is $M = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ with eigenvalues φ (expanding) and ψ (contracting, $|\psi| < 1$).*

Definition 2.4 (Clean Operation). *$\text{clean}_r(a, b) = \text{div}_Y \circ \text{mul}_Y^r(a, b)$. Its eigenvalues are φ^{r-1} and ψ^{r-1} . For $r = 3$, the ψ -component is multiplied by $\psi^2 \approx 0.382$ —exponential attenuation with zero multiplicative depth cost.*

3 Noise Management for Fully Homomorphic Encryption

3.1 Construction

We implement the φ -extension on top of CKKS [5] in OpenFHE [6]. Each element of R_φ is represented as a pair of CKKS ciphertexts $(\mathbf{a}, \mathbf{b})$.

Ring multiplication follows from $(a_1 + b_1Y)(a_2 + b_2Y) = a_1a_2 + b_1b_2 + (a_1b_2 + a_2b_1 + b_1b_2)Y$, requiring 4 **EvalMult** and 3 **EvalAdd** operations. This is comparable in cost to a single CKKS ciphertext multiplication after accounting for the dual representation.

3.2 Noise Dynamics

Let $\mathbf{c} = (\mathbf{a}, \mathbf{b})$ encrypt message m . The noise in each reality is:

$$\eta_\varphi = |\text{Dec}(\mathbf{a}) + \varphi \cdot \text{Dec}(\mathbf{b}) - m| \quad (2)$$

$$\eta_\psi = |\text{Dec}(\mathbf{a}) + \psi \cdot \text{Dec}(\mathbf{b}) - m| \quad (3)$$

The clean operation (ratio 3:1) attenuates η_ψ by factor $\psi^2 \approx 0.382$ while amplifying η_φ by $\varphi^2 \approx 2.618$. We compensate for the amplification through pre-scaling of multiplication operands by $\varphi^{-2/3} \approx 0.726$.

Important correction: In an earlier version of this work, we described φ -error growth as “linear.” This was incorrect. Re-analysis of our data shows exponential growth with a very small base:

$$\eta_\varphi(n) \approx \eta_\varphi(0) \cdot (1.00075)^n \quad (4)$$

where n is the number of multiplications. The growth rate is approximately 0.075% per multiplication. This is slow enough to be practical—projecting to 1% error yields $\sim 14,000$ multiplications—but it is exponential, not linear. We thank the anonymous reviewer who caught this error.

3.3 Bootstrap Integration

The clean operation attenuates noise but does not refresh the CKKS modulus levels. For extended computation, we periodically apply standard CKKS bootstrapping followed by 2 recovery clean cycles. The bootstrap injects fresh noise into both realities; the recovery cleans attenuate the ψ -component back to baseline.

3.4 Cross-Library Portability

The φ -transform requires only three primitives: **EvalAdd**, **EvalMult**, and encryption of zero. These are available in all Ring-LWE-based FHE libraries. We verified API-level compatibility across 11 libraries (OpenFHE, SEAL, Helib, Lattigo, TFHE, FHEW, TenSEAL, Pyfhel, Concrete, TFHE-rs, PALISADE). We note this verification checks that the required operations exist, not that the φ -transform produces correct results under each library’s specific encoding and noise model.

3.5 Experimental Results

All experiments on AMD Ryzen 5 2600, 16GB RAM, WSL2 on Windows 11, OpenFHE v1.2, CKKS RingDim=8192.

Table 1: 100-Bootstrap Run (RingDim=8192, Clean=3:1)

Bootstrap	Mults	φ -error	ψ -noise	Status
0	60	1.87×10^{-9}	1.83×10^{-11}	OK
10	690	2.19×10^{-8}	8.49×10^{-12}	OK
20	1320	4.21×10^{-8}	2.17×10^{-11}	OK
30	1950	6.16×10^{-8}	6.93×10^{-12}	OK
40	2580	8.13×10^{-8}	9.11×10^{-12}	OK
50	3210	1.01×10^{-7}	9.23×10^{-12}	OK
60	3840	1.21×10^{-7}	8.93×10^{-12}	OK
70	4470	1.41×10^{-7}	5.50×10^{-12}	OK
80	5100	1.60×10^{-7}	2.13×10^{-12}	OK
90	5730	1.80×10^{-7}	4.51×10^{-12}	OK
99	6297	1.98×10^{-7}	1.15×10^{-11}	OK

Error growth analysis:

- Start (60 mults): 1.87×10^{-9}
- End (6297 mults): 1.98×10^{-7}
- Growth factor over 6237 mults: $\sim 106\times$
- Per-multiplication factor: $106^{1/6237} \approx 1.00075$ (0.075% growth per mult)
- Projected to 1% error (10^{-2}): $\sim 14,000$ multiplications

ψ -noise: Remains bounded in $[10^{-12}, 10^{-11}]$, showing no accumulation trend. Bootstrap noise spikes recover within 2 clean cycles.

Fibonacci depth compression (scope clarification): Zeckendorf’s theorem [7] enables computing x^N (exponentiation of a single value) with $O(\log N)$ rather than $O(N)$ multiplications. For $N = 10^9$, this reduces 10^9 multiplications to 56. We note this technique applies specifically to exponentiation chains, not to arbitrary sequences of different multiplications.

4 Dual-Reality Program Encoding

4.1 Concept

The CRT decomposition means any computation in R_φ produces two outputs simultaneously. We can encode two functionally equivalent representations of a computation such that:

1. Both representations produce the correct output for all inputs
2. An observer without the secret key cannot determine which representation was the intended one

What this is: A form of plausible deniability for program representation. Given two algebraically equivalent ways to express a computation (e.g., factored vs. expanded polynomial evaluation), our encoding hides which representation was used.

What this is NOT: Indistinguishability obfuscation (iO) in the sense of Barak et al. [8] or Garg et al. [9]. We do not claim to hide arbitrary circuits, to provide virtual black-box obfuscation, or to satisfy the strong definitions of cryptographic iO. The term “dual-reality program encoding” more accurately describes our contribution.

4.2 The Fibonacci Circle Mapping

To prevent an adversary from distinguishing representations by their intermediate computation traces, we map Fibonacci ratios $F(n)/F(n+1)$ to points on the unit circle:

$$z_n = \exp\left(2\pi i \cdot \frac{F(n)}{F(n+1)}\right) \quad (5)$$

Both normal and reverse sequences produce symmetric trajectories converging to $\exp(2\pi i/\varphi)$ with identical total angular distance. The adversary sees two circle paths with equal metrics and cannot identify which corresponds to which representation.

4.3 Adversary Model

The adversary in our experiment receives:

- The obfuscated program (a φ / ψ -encoded ciphertext pair)
- The ability to query the program on arbitrary inputs
- The two evaluation outputs (φ -reality and ψ -reality values)
- Knowledge of the obfuscation algorithm (Kerckhoffs’s principle)

The adversary does NOT receive the secret key, and therefore cannot determine which evaluation corresponds to which reality. The adversary’s task is to identify whether the φ -reality output came from the first or second representation, given both outputs and unlimited queries.

4.4 Adversary Game Results

1,000 trials. Adversary must identify which representation was used.

Table 2: Adversary Distinguishability

Method	Success Rate	Baseline
Direct value sequences	100.0%	50%
Convergent ratios	51.2%	50%
Circle mapping	51.2%	50%
Phoenix (unified)	51.3%	50%

For $n = 1000$, 95% CI for random guessing: $50\% \pm 3.1\%$. Observed 51.2% is within this interval ($z \approx 0.76$, $p \approx 0.45$, two-tailed). We cannot reject the null hypothesis of random guessing.

Limitation: This test only shows a particular simple adversary cannot distinguish the representations. It does not prove security against all polynomial-time adversaries, as required for cryptographic iO.

4.5 Scope of Applicability

Our encoding applies to algebraically equivalent representations of polynomial evaluation (e.g., factored vs. expanded forms, different parenthesizations). For a degree- d polynomial, there are Catalan-number many distinct evaluation trees, all computing the same function. After padding to equal gate counts, any pair can be encoded.

This is a restricted class—much smaller than general circuits. An adversary who knows the polynomial being computed can evaluate it directly, bypassing the encoding entirely. The technique is useful when the polynomial itself is public but the evaluation strategy is proprietary (e.g., optimized evaluation algorithms, hardware-specific implementations).

5 Compact Key Encapsulation

5.1 Construction

We apply the φ -extension to compress Ring-LWE public keys. Instead of storing the full polynomial $T = A \cdot S + E \in R_q$, we store only $\text{eval}_\varphi(T)$ and $\text{eval}_\psi(T)$. These two evaluations determine T uniquely (by CRT) but recovering them without the secret requires solving Ring-LWE.

5.2 Assumptions and Caveats

Our construction differs from NIST-standardized KEMs in two important ways:

1. **Ring-LWE vs. Module-LWE.** Kyber [11] uses Module-LWE, which provides tighter worst-case to average-case reductions than Ring-LWE [12, 13]. Ring-LWE’s additional algebraic structure could potentially admit specialized attacks. We use Ring-LWE because it enables smaller parameters; the trade-off is a less conservative security assumption.
2. **Fixed public matrix A .** Unlike Kyber, which derives a fresh A per key, we use a single fixed A for all users. If A is found to have a trapdoor, all keys are compromised simultaneously. Standard RLWE security proofs assume fresh random A per instance. We acknowledge this is non-standard and requires formal analysis.

Important: The size comparison to Kyber-1024 in Table 3 is provided for scale reference only. It is **not** a claim of equivalent security. Our 192-byte construction uses weaker assumptions to achieve smaller size. A fair comparison would require both schemes to operate under identical assumptions, which they do not.

Table 3: Experimental KEM Variants

Variant	SK	PK	CT	Total	Assumption
φ -KEM QR	16B	32B	32B	80B	RLWE, fixed A
φ -KEM v5	32B	64B	32B	128B	RLWE, fixed A
φ -KEM Ultra	128B	640B	1024B	1792B	RLWE, fixed A
φ -KEM L5	64B	64B	64B	192B	RLWE, fixed A
Kyber-512	1632B	800B	768B	3200B	MLWE, fresh A
Kyber-1024	3168B	1568B	1568B	6304B	MLWE, fresh A

5.3 Experimental Results

All variants tested with 20–1,000 encaps/decaps cycles, 100% correctness, 100% tamper detection. Code available in repository.

6 Related Work

6.1 FHE Noise Management

Gentry’s bootstrapping [1] refreshes ciphertexts by evaluating decryption homomorphically. Scale-invariant schemes [2] and CKKS rescaling [5] reduce noise growth but do not eliminate it. Our clean operation is orthogonal: it provides additional attenuation using only addition operations, complementing existing techniques.

6.2 Program Obfuscation

General-purpose iO was constructed by Garg et al. [9] using multilinear maps. Subsequent attacks [14, 15] have made practical iO difficult. Our encoding is far more limited in scope and should not be compared to these works. We provide a practical tool for a specific use case (hiding evaluation strategy), not a general solution.

6.3 Post-Quantum KEM

NIST selected Kyber [11] (Module-LWE) as the primary KEM standard. Our construction explores a different point in the design space (Ring-LWE with aggressive compression), trading security margin for size. Similar compression ideas appear in lattice signatures [16].

7 Limitations

We document limitations honestly:

1. **No third-party verification.** All results are author-reported. Independent reproduction is essential.
2. **Exponential error growth.** φ -error grows exponentially with base ~ 1.00075 , not linearly as we initially claimed. This limits practical depth to $\sim 14,000$ multiplications at 1% error, not “unlimited.”
3. **Dual-reality encoding is not iO.** Our construction provides plausible deniability for program representation, not general indistinguishability obfuscation. We have corrected terminology throughout.
4. **KEM uses weaker assumptions.** Ring-LWE with fixed matrix A is less studied than Module-LWE with fresh A . Size comparisons to Kyber are illustrative, not claims of equivalent security.
5. **TOY security parameters.** FHE benchmarks at RingDim=4096/8192. Production security requires larger dimensions.
6. **Cross-library verification is API-level only.** We verified that required operations exist across libraries, not full correctness of φ -transform under each library’s encoding.
7. **Not constant-time.** Current implementations are not hardened against side-channel attacks.

8. **Bootstrap latency.** ~ 40 seconds per bootstrap on consumer CPU.
9. **Zeckendorf compression scope.** Applies to exponentiation chains, not arbitrary sequential multiplications.

8 Conclusion

We have explored the φ -extension ring $R[X]/(X^2 - X - 1)$ as a primitive for lattice-based cryptography. Our exploration yielded:

- A noise attenuation technique for FHE that uses zero multiplication depth operations, enabling extended computation chains (demonstrated to 6,300 multiplications, projected to $\sim 14,000$ practical limit)
- A dual-reality program encoding that provides plausible deniability for evaluation strategy (not general iO)
- A compact KEM construction trading assumption strength for key size (192 bytes with RLWE + fixed A)

We have corrected our initial error regarding linear vs. exponential growth, clarified the scope of our program encoding relative to standard iO definitions, and explicitly documented the weaker assumptions in our KEM. We believe the φ -extension ring is a genuinely interesting algebraic structure that warrants further study, particularly its asymmetric eigenvalue dynamics.

All code is publicly available. We welcome independent verification, critique, and collaboration.

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